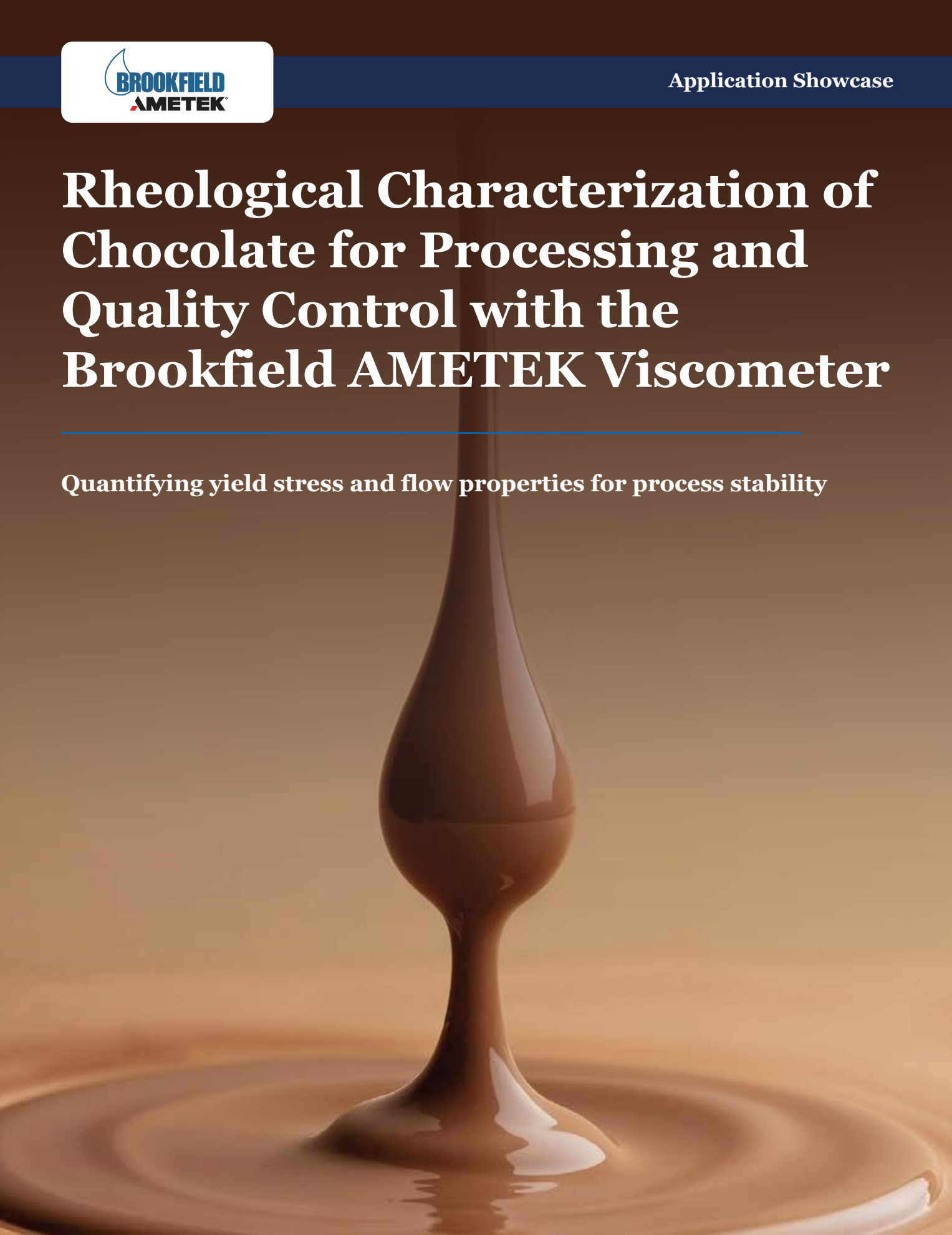


Rheological Characterization of Chocolate for Processing and Quality Control with the Brookfield AMETEK Viscometer

Quantifying yield stress and flow properties for process stability



The flow behaviour of chocolate plays a critical role in manufacturing operations such as pumping, mixing, enrobing, and molding. Chocolate exhibits non-Newtonian, shear-thinning behaviour and requires precise temperature control during measurements to obtain reliable viscosity data. This application demonstrates the rheological evaluation of dark chocolate at 40 °C using a rotational viscometer with a small sample adapter.

Keywords or phrases:

Chocolate, rheology, shear-thinning, Casson model, yield stress, viscosity, process control, Ametek Brookfield

INTRODUCTION

The processability of chocolate depends strongly on rheological properties such as viscosity, yield stress, and shear response.

These parameters influence:

- ▶ Pumping and pipeline transport
- ▶ Enrobing and coating thickness
- ▶ Mold filling and air release
- ▶ Final texture and mouthfeel

Chocolate behaves as structured suspension of cocoa solids, sugar particles, and fat, exhibiting shear-thinning behaviour; viscosity decreases as shear rate increases. Accurate rheological characterization therefore requires measurements across multiple shear conditions and strict temperature control, as viscosity is highly temperature dependent

MATERIALS AND EQUIPMENT

Instrument: Viscometer or Rheometer

Torque Range: HA

Spindle: SC4-27 Spindle with SC4-13RPY Sample Chamber

Accessories: Small Sample Adapter,
TC-550SD Programmable Temperature Bath

Speed: 5, 10, 20, 50 and 100 RPM

Temperature: 40°C

TEST METHOD

The Small Sample Adapter™ may be used with various Brookfield Viscometers or Rheometers. In our example, we used a Brookfield HADV3T, with RheocalcT™ software for automated instrument control and data acquisition.

Our test temperature of 40°C was maintained by connecting the Small Sample Adapter™ water jacket to a Brookfield TC-550SD Programmable Refrigerated Bath. Representative data from the analyses are shown in Figures 1 and 2.

Figure 1 demonstrates that the viscosity of the dark chocolate decreased as the shear rate increased. This behaviour is known as “shear- thinning”.

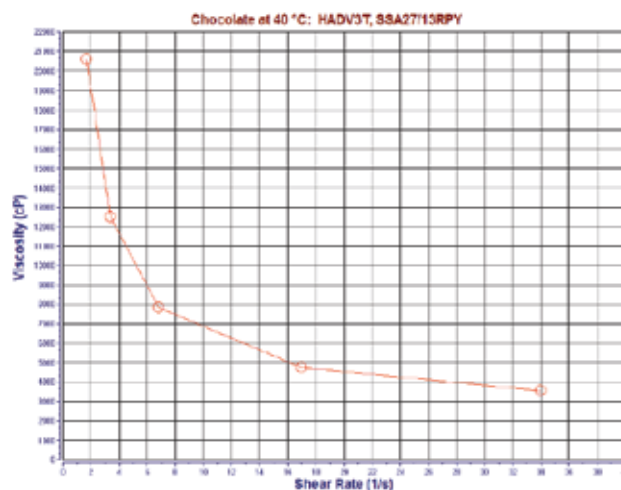


Figure 1: Dark Chocolate Viscosity at 40°C

Figure 2 illustrates a good fit of the NCA/CMA Casson model to the raw data. The “CoF”, the Confidence of Fit (%), of the raw data to the fitted curve, is nearly 100% signifying an excellent fit. As given in Figure 2, generated in RheocalcT™, the plastic viscosity is 1241 cP and the yield stress is 129 dyn/cm². The chocolate was tested according to specifications provided by the National Confectioners Association and Chocolate Manufacturers Association.

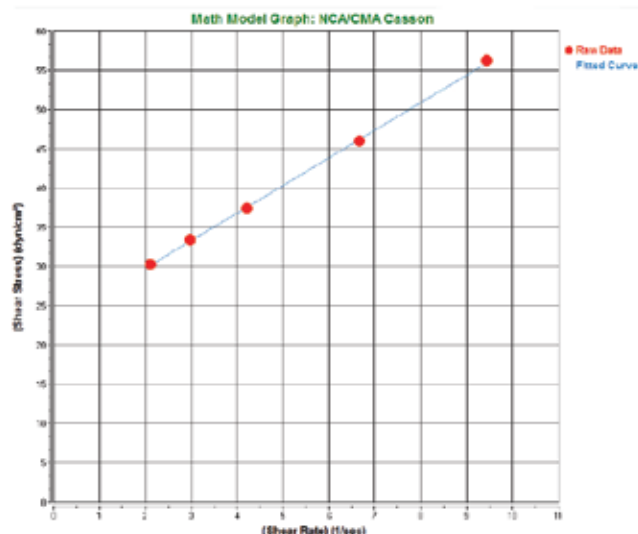


Figure 2: NCA/CMA Casson Plot of Chocolate Rheology at 40°C

Some chocolates are tested at different shear rates, to ensure on-scale data. 1, 2.5, 5, 10, 20 and 50 RPM are appropriate in some cases, for example.

The Degrees MacMichael values may be calculated from Brookfield data at 20 RPM. The Torque, %, reading at 20 RPM, obtained by measuring chocolate with the HA spring torque range instrument using the SC4-27 spindle in the SC4-13R or SC4-13RP chamber, is multiplied by 3.40. The above example has a torque of 31.4% at 20 RPM – corresponding to a viscosity of 7850 cP at a shear rate of 6.8 s⁻¹ or 6.8 1/s. Therefore, (31.4%)x(3.40) = 106.8 degrees MacMichael.

CONCLUSION

Chocolate exhibits complex shear-dependent flow behaviour that must be carefully characterized to ensure manufacturing efficiency and product consistency. Rotational viscosity measurement under controlled temperature conditions, combined with Casson model analysis, provides reliable determination of plastic viscosity and yield stress. These parameters enable manufacturers to maintain process control, meet industry specifications, and deliver consistent product quality.

DV3T™ Viscometer

The all-in-one tool for measuring viscosity and yield stress



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